

Equivariant-Decomposed-Residual (EDR) Networks

Structurally-Constrained Neural Corrections for Robot Inverse Dynamics

Gravity

Inertia

Coriolis

Friction

SHRI Lab

Smart Healthcare & Robotics Interfacing Laboratory

April 21, 2026

Motivation: Why Structured Corrections?

The Inverse Dynamics Problem

Given trajectory $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$, predict joint torques $\boldsymbol{\tau}$.

Nominal Physics (RNEA):

$$\boldsymbol{\tau}_{\text{nom}} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \boldsymbol{\tau}_f(\dot{\mathbf{q}})$$

... but suffers from *model–reality gaps*:

- ✗ Inaccurate inertial parameters
- ✗ Unmodelled cable / gear friction
- ✗ Joint flexibility & backlash
- ✗ Simplified geometry (CAD $\neq$ reality)

Three Modelling Paradigms

Black-Box NN

High capacity, no physics, > \$200k params

↓ no physics

Pure Physics (RNEA)

Perfect structure, wrong parameters

↓ augment

EDR (This Work)

Physics backbone + structured δ -corrections, $\sim$ \$5k params

Core Idea: Learn a *small, structured correction* $\delta\boldsymbol{\tau}$ to each physics component—gravity, inertia, Coriolis, friction—while **enforcing physical constraints by architecture**, not by loss penalties.

EDR: Mathematical Formulation

Equivariant-Decomposed-Residual decomposition of predicted torque:

$$\hat{\tau} = [\mathbf{M} + \delta\mathbf{M}] \ddot{\mathbf{q}} + [\boldsymbol{\tau}_C + \delta\mathbf{C}\dot{\mathbf{q}}] + [\mathbf{g} + \delta\mathbf{g}] + [\boldsymbol{\tau}_f + \delta\boldsymbol{\tau}_f]$$

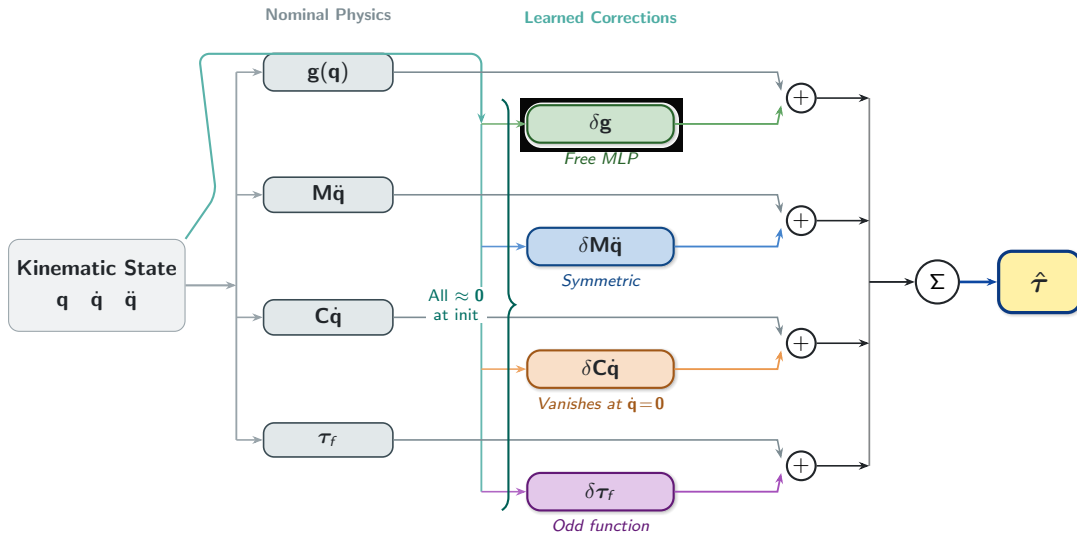
Four Correction Networks

Network	Form	Constraint
$\delta\mathbf{g}(\mathbf{q})$	$\text{MLP}_{\mathbf{g}}(\mathbf{q}_{\text{aug}})$	None (free)
$\delta\mathbf{M}(\mathbf{q})$	Symmetric scatter	$\delta\mathbf{M} = \delta\mathbf{M}^\top$
$\delta\mathbf{C}\dot{\mathbf{q}}$	$\dot{\mathbf{q}} \odot \text{MLP}_C$	$\rightarrow \mathbf{0}$ when $\dot{\mathbf{q}} = \mathbf{0}$
$\delta\boldsymbol{\tau}_f(\dot{\mathbf{q}})$	$\dot{\mathbf{q}} \odot h_\phi(\dot{\mathbf{q}})$	Odd: $f(-x) = -f(x)$

Design Principles

- ✓ **Zero-initialised:** all δ -nets output $\approx \mathbf{0} \Rightarrow \hat{\tau} = \boldsymbol{\tau}_{\text{phys}}$ before training.
- ✓ **Equivariant structure:** each residual transforms with its physics term—e.g., friction is odd in $\dot{\mathbf{q}}$ and velocity-gated terms vanish at rest.
- ✓ **Decomposed residuals:** each network targets one physics component $\Rightarrow$ interpretable.

EDR: Complete Architecture



Gravity Correction Network $\delta \mathbf{g}(\mathbf{q})$

Physical Motivation

Gravity torque derives from potential energy $V(\mathbf{q})$ and depends *only on configuration* $\mathbf{q}$ —via trig functions of joint angles.

Mathematical Form

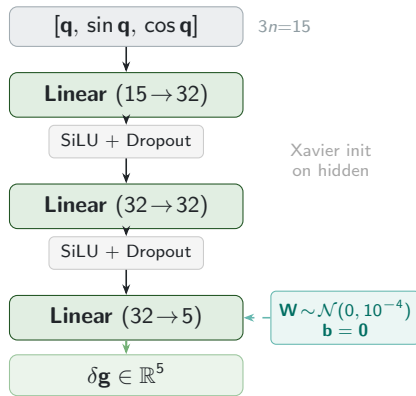
$$\delta \mathbf{g}(\mathbf{q}) = \text{MLP}_g(\mathbf{q}_{\text{norm}}, \sin \mathbf{q}_{\text{raw}}, \cos \mathbf{q}_{\text{raw}}) \in \mathbb{R}^n$$

Why Trig Augmentation?

- Gravity depends on $\sin / \cos$ of joint angles
- Avoids MLP re-discovering trig structure
- Raw angles: $\mathbf{q}_{\text{raw}} = \mathbf{q}_{\text{norm}} \cdot \sigma_q + \mu_q$

Constraint: None—free additive vector.

Network Architecture



$\sim \$1,700$ trainable parameters

Inertia Correction Network $\delta\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}$

Physical Motivation

The inertia matrix $\mathbf{M}(\mathbf{q})$ must be *symmetric*. The correction $\delta\mathbf{M}$ must preserve this.

Symmetric-by-Construction

MLP outputs $\frac{n(n+1)}{2} = 15$ values (for $n=5$), scattered into both triangles:

$$\delta M_{ij} = \delta M_{ji} \quad \forall i, j$$

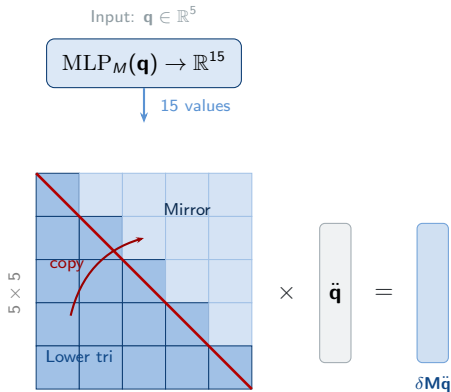
Torque contribution:

$$\delta\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} = \text{scatter}(\text{MLP}_M(\mathbf{q})) \ddot{\mathbf{q}}$$

Why Not Cholesky ($\mathbf{LL}^\top$)?

- $\mathbf{LL}^\top$ forces positive-definiteness
- Corrections can be positive or negative
- Direct scatter avoids gradient deadlock

Symmetric Scatter Construction



~1,400 params — Symmetry exact $\forall \theta$

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Coriolis Correction Network $\delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$

Physical Constraint

Coriolis/centripetal torques vanish when stationary:

$$\mathbf{C}(\mathbf{q}, \mathbf{0})\mathbf{0} = \mathbf{0} \implies \delta\mathbf{C}(\mathbf{q}, \mathbf{0})\mathbf{0} = \mathbf{0}$$

Element-wise Velocity Gating

$$\delta\mathbf{C}\dot{\mathbf{q}} = \dot{\mathbf{q}} \odot \text{MLP}_C(\text{features})$$

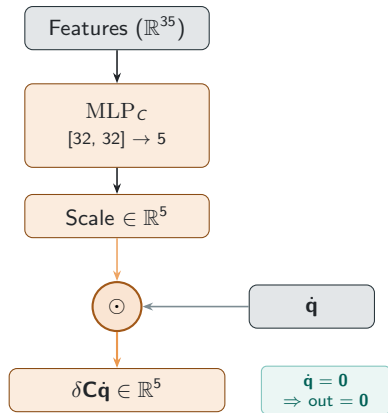
Since $\dot{\mathbf{q}} = \mathbf{0}$ zeroes the product regardless of MLP output, the constraint is **exact**.

Physics-Informed Features

$$\text{feat} = [\mathbf{q}, \sin \mathbf{q}, \cos \mathbf{q}, \dot{\mathbf{q}}, \dot{q}_i \dot{q}_j]$$

Dimension: $4n + \frac{n(n+1)}{2} = 35$ for $n=5$

Gating Mechanism



Friction Correction Network $\delta\tau_f(\dot{\mathbf{q}})$

Physical Constraint

Friction reverses sign with motion:

$$\tau_f(-\dot{\mathbf{q}}) = -\tau_f(\dot{\mathbf{q}}) \text{ (odd function)}$$

Product Trick Construction

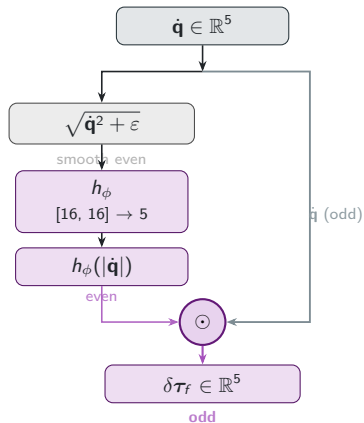
$$\delta\tau_f(\dot{\mathbf{q}}) = \underbrace{\dot{\mathbf{q}}}_{\text{odd}} \odot \underbrace{h_\phi(|\dot{\mathbf{q}}|)}_{\text{even}}$$

Proof of Oddness

$$\begin{aligned}\delta\tau_f(-\dot{\mathbf{q}}) &= (-\dot{\mathbf{q}}) \odot h_\phi(|\dot{\mathbf{q}}|) \\ &= -\delta\tau_f(\dot{\mathbf{q}}) \quad \checkmark\end{aligned}$$

Smooth Absolute Value: $|\dot{q}| \approx \sqrt{\dot{q}^2 + \varepsilon}$
($\varepsilon = 10^{-6}$, differentiable at $\dot{q} = 0$)

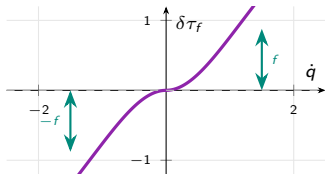
Network Data Flow



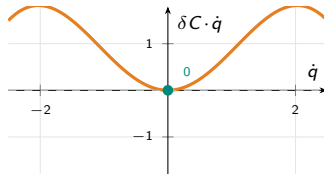
Correction Response Characteristics

Illustrative response shapes enforced by architectural constraints (before training):

Friction (Odd)

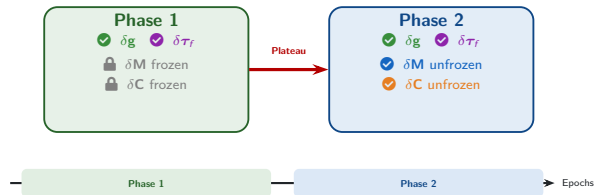


Coriolis (Vanishes at 0)



Property	Mechanism	Guarantee
$\delta \tau_f(-\dot{q}) = -\delta \tau_f(\dot{q})$	$\dot{q} \odot h(\dot{q})$: odd $\times$ even = odd	Exact $\forall \theta$
$\delta C \cdot \dot{q} _{\dot{q}=0} = 0$	$\dot{q} \odot \text{MLP}$: zero factor	Exact $\forall \theta$
$\delta \mathbf{M} = \delta \mathbf{M}^\top$	Symmetric scatter of $\frac{n(n+1)}{2}$ entries	Exact $\forall \theta$

Two-Phase Curriculum Learning



Why Two Phases?

- ➔ Phase 1 clears largest residuals (gravity & friction)
- ➔ Phase 2 fine-tunes coupling terms
- ➔ Prevents δM absorbing gravity errors

Adaptive Phase Transition

Triggered when validation RMSE plateaus:

$$\frac{\text{RMSE}_{t-W} - \text{RMSE}_t}{\text{RMSE}_{t-W}} < \theta \Rightarrow \text{Phase 2}$$

Param	Value	Role
Window W	5	Smoothing
Threshold θ	0.5%	Sensitivity
Min epoch	3	Noise guard
Max epoch	25	Safety cap

On Transition:

1. Unfreeze δM , δC
2. Equalise learning rates
3. Clear stale Adam state

Training Loss Function

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{stab}} \mathcal{L}_{\text{stab}}$$



Prediction Error

Match real torque data

$$\mathcal{L}_{\text{data}} = \frac{1}{Bn} \sum_{b,j} (\hat{\tau}_{b,j} - \tau_{b,j})^2$$



Keep Corrections Small

Trust physics unless data disagrees

$$\mathcal{L}_{\text{reg}} = \|\delta \mathbf{g}\|^2 + \frac{\|\delta \mathbf{M}\|^2}{n} + \|\delta \mathbf{C} \dot{\mathbf{q}}\|^2 + \|\delta \boldsymbol{\tau}_f\|^2$$



Stability (optional)

No spontaneous energy

$$\mathcal{L}_{\text{stab}} = \|\mathbf{S} + \mathbf{S}^T\|^2$$

Term	Weight	What It Does
Prediction error	0.7	Match measured torques
Correction size	0.25	Do not over-correct physics
Stability	0.05	Keep model physically safe

Weight Initialisation: Starting at Physics Baseline

The Zero-Init Dilemma

✗ Exactly Zero Weights

$\partial L / \partial \text{hidden} = \mathbf{W}_{\text{last}}^T \times \text{upstream} = 0$
 $\Rightarrow$ Gradient deadlock

Solution

✓ Near-Zero ($\sigma = 10^{-4}$)

Output $\sim \mathcal{O}(10^{-3})$ —negligible vs physics $\sim \mathcal{O}(1)$
Gradients alive from step 1

Initialisation Recipe

Layer	Weights	Biases
Hidden	Xavier normal	0
Final	$\mathcal{N}(0, 10^{-4})$	0

Consequence at Epoch 0

At Construction

$\delta \mathbf{g} \approx 0, \delta \mathbf{M} \approx 0$
 $\delta \mathbf{C}\dot{\mathbf{q}} \approx 0, \delta \boldsymbol{\tau}_f \approx 0$

therefore

$\hat{\boldsymbol{\tau}} = \boldsymbol{\tau}_{\text{phys}}$

Model starts at nominal physics

Training can only improve

Verified: $\max |\hat{\boldsymbol{\tau}} - \boldsymbol{\tau}_{\text{phys}}| < 5 \times 10^{-3}$ across 512 random samples.

Optimizer & Regularisation Strategy

AdamW with Two Parameter Groups

Group 0: $\delta \mathbf{g} + \delta \boldsymbol{\tau}_f$

LR = 3×10^{-4} (full)

Group 1: $\delta \mathbf{M} + \delta \mathbf{C}$

LR = frozen_lr_ratio $\times$ LR

Non-zero LR maintains Adam buffers

Learning Rate Schedule

Cosine annealing with warm restarts:

- $T_0 = 15$ epochs, $\eta_{\min} = 3 \times 10^{-6}$
- Fixed cycle length ($T_{\text{mult}} = 1$)

Multi-Layer Regularisation

 **Weight Decay** 2×10^{-3}

 **Correction Reg** 5×10^{-2}

 **Dropout** 0.25

 **Feature Noise** $\sigma = 0.02$

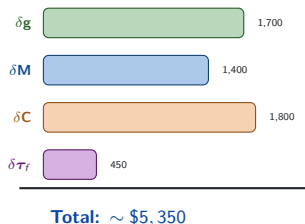
 **Early Stopping** pat = 25

Prevents
overfitting

Design philosophy: Heavy regularisation ensures train and validation loss descend in lockstep—no post-peak overfitting observed.

Parameter Summary & Model Comparison

EDR Parameter Breakdown ($n = 5$ joints)



Hidden: [32,32] for $g/M/C$; [16,16] for friction. SiLU activation.

Comparison with Other Approaches

Model	Params	Physics?
Black-Box FNN	> \$200k	✗ None
Residual/Regularized NN	~ \$150k	⚠ Additive
EDR (ours)	~ \$5k	✓ Structured

EDR Advantages

- ✓ ~40× fewer parameters than black-box
- ✓ **Interpretable**—each correction is physically meaningful
- ✓ **Guaranteed** symmetry, oddness, vanishing
- ✓ **Safe init**—starts at physics baseline
- ✓ **Fast convergence**—physics backbone is a strong prior

Summary: EDR Networks



Physics-Informed

4 decomposed corrections
targeting specific terms



Constrained

Symmetry, oddness,
vanishing by architecture



Lightweight

$\sim 5k$ params total
($\sim 40\times$ smaller)



Safe & Verified

Zero-init baseline
35+ structural tests

EDR = strong physics prior + minimal corrections + guaranteed structural properties

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